

Introduction to **Pharmacology**

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PHARMACOLOGY

The science that study the **interactions** between **drugs** and **living systems**.

It include the study of:

- ❖ Effects of **drugs** on human body = (**Pharmacodynamics**)
- ❖ Effects of **human body** on drugs = (**Pharmacokinetics**)

DRUG

Any substance that make a **change in biologic function** when administered to a living organism, through its **chemical actions**.

The **change in biologic function** is referred to: **biological effects**

The drug is usually crude chemical compound.

MEDICINE

The **final dosage form** of the drug Like (**tablet, capsule, ampule, syrupetc.**) that can be administered to the body, it contain the drug (active ingredient) and other **excipients** (inert ingredients) like sugar, starch. ...etc.

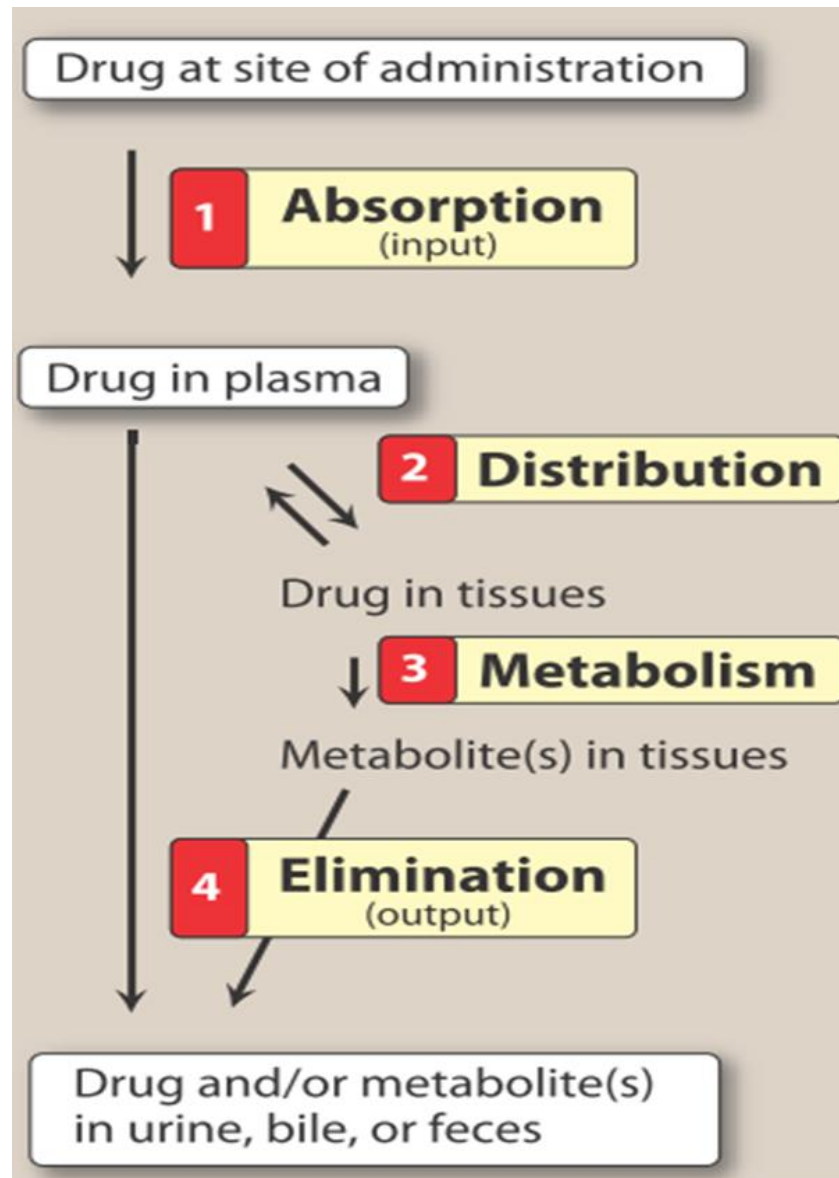
Pharmacodynamics

Effects of drugs on the body:

- **Therapeutic effect** (effect of the drug on the therapeutic target)
- **Side effect or adverse effect** (effect of the drug on other targets). Some time the side effects have clinical benefit, like the sedative effect of some antihistamines.
- **Toxic effect** (harmful effects, always unwanted and should be avoided during therapy)
- **Lethal effect** (death)

The science that study the harmful effects of drugs known as : **Toxicology**

Pharmacokinetics:



Pharmacokinetics:

Terms:

Dose

Dosage form

Route of drug administration

Bioavailability

Half life ($t_{1/2}$)

Drug elimination

Drug excretion

Drug metabolism

PHARMACOLOGY

For each drug (or group of drugs) studied, you should know the followings:

- Classification (drug class)
- Mechanism of action
- Pharmacokinetics (administration, $t_{1/2}$, metabolism,etc.)
- Pharmacological effects
- Clinical use (indications)
- Adverse effects and toxicity
- Contraindications and precautions
- Drug interactions
- Any specific information related to drug use.

REFERENCES

- **Lippincott's** Illustrated Reviews: Pharmacology
- **Katzung's** Basic & Clinical Pharmacology
- **Rang and Dale's** Pharmacology

THANK YOU

